COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE212 - Automating Server Management | **SECTION:** | CPE31S2 |
| **DATE PERFORMED:** | 09/02/2026 | **DATE SUBMITTED:** | 09/03/2026 |
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**ACTIVITY NO. 6** **Targeting Specific Nodes**

## 1. PROCEDURE OUTPUT

### Task 1: Targeting Specific Nodes

1. Create a new playbook and named it site.yml. Follow the commands as shown in the image below. Make sure to save the file and exit.

```
 GNU nano 8.4                                                site.yml
---

- hosts: all
  become: true
  tasks:

    - name: Install apache and php for ubuntu servers
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == 'Ubuntu"

    - name: install apache and php for CentOS servers
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"
```

2. Edit the inventory file. Remove the variables we put in our last activity and group according to the image shown below:

```
 GNU nano 8.4                                                    inventory.yaml
all:
  vars:
    ansible_user: charles-vence-espinas
  children:

      web_servers:
        hosts:
          server1:
            ansible_host: 192.168.56.102
            os_type: Ubuntu

      db_servers:
        hosts:
          server2:
            ansible_host: 192.168.56.103
            os_type: Ubuntu

      file_servers:
        hosts:
          centOS:
            ansible_host: 192.168.56.104
            os_type: CentOS
```

Make sure to save the file and exit.

Right now, we have created groups in our inventory file and put each server in its own group. In other cases, you can have a server be a member of multiple groups, for example you have a test server that is also a web server.

- So as you can see, I assign the server 1 which is ubuntu for webserver, server 2 which is ubuntu for db server and cent os for file servers.

3. Edit the site.yml by following the image below:

```
 GNU nano 8.4                                                      site.yml
---

- hosts: all
  become: true
  pre_tasks:

    - name: Install Updates (CentOS)
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: Install updates (Ubuntu)
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  tasks:

    - name: Install apache and php for ubuntu servers
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"
```

```
  - name: install apache and php for CentOS servers
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"
```

Make sure to save the file and exit.

The pre-tasks command tells the ansible to run it before any other thing. In the pre-tasks, CentOS will install updates while Ubuntu will upgrade its distribution package. This will run before running the second play, which is targeted at web_servers. In the second play, apache and php will be installed on both Ubuntu servers and CentOS servers.

Run the site.yml file and describe the result.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml site.yml --ask-become-pass
BECOME password:

PLAY [all] ***********************************************************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server2]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for
more information.
ok: [centOS]

TASK [Install Updates (CentOS)] **************************************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] **************************************************************************************************************************
skipping: [centOS]
ok: [server2]
ok: [server1]

PLAY [web_servers] ***************************************************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************************************************
ok: [server1]

TASK [Install apache and php for ubuntu servers] *********************************************************************************************************
changed: [server1]
```

```
TASK [install apache and php for CentOS servers] *********************************************************************************************************
skipping: [server1]

PLAY RECAP ***********************************************************************************************************************************************
centOS                     : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server1                    : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

- It is successful because the playbook is designed for the installation of both os that I have which is the ubuntu os and cent os.

4. Let's try to edit again the site.yml file. This time, we are going to add plays targeting the other servers. This time we target the db_servers by adding it on the current site.yml. Below is an example: (Note add this at the end of the playbooks from task 1.3.

```
 GNU nano 8.4                                    site.yml
---

- hosts: all
  become: true
  pre_tasks:

    - name: Install Updates (CentOS)
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: Install updates (Ubuntu)
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  tasks:

    - name: Install apache and php for ubuntu servers
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"
```

```
    - name: install apache and php for CentOS servers
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"

- hosts: db_servers
  become: true
  tasks:

    - name: Install mariadb package (CentOS)
      yum:
        name: mariadb-server
        state: latest
      when: ansible_distribution == "CentOS"

    - name: "Mariadb- Restarting/Enabling"
      service:
        name: mariadb
        state: restarted
        enabled: true

    - name: install mariadb package (Ubuntu)
      apt:
        name: mariadb-server
        state: latest
      when: ansible_distribution == "Ubuntu"
```

Make sure to save the file and exit.
Run the site.yml file and describe the result.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml site.yml --ask-become-pass
BECOME password:

PLAY [all] ***********************************************************************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for
more information.
ok: [centOS]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server2]
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server1]

TASK [Install Updates (CentOS)] **************************************************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] **************************************************************************************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [web_servers] ***************************************************************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************************************************************
ok: [server1]

TASK [Install apache and php for ubuntu servers] *********************************************************************************************************************
ok: [server1]
```

```
TASK [install apache and php for CentOS servers] *********************************************************************************************************************
skipping: [server1]

PLAY [db_servers] ****************************************************************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************************************************************
ok: [server2]

TASK [Install mariadb package (CentOS)] ******************************************************************************************************************************
skipping: [server2]

TASK [Mariadb- Restarting/Enabling] **********************************************************************************************************************************
[ERROR]: Task failed: Module failed: Could not find the requested service mariadb: host
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/site.yml:50:7

48       when: ansible_distribution == "CentOS"
49
50     - name: "Mariadb- Restarting/Enabling"
         ^ column 7

fatal: [server2]: FAILED! => {"changed": false, "msg": "Could not find the requested service mariadb: host"}

PLAY RECAP ***********************************************************************************************************************************************************
centOS                     : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server2                    : ok=3    changed=0    unreachable=0    failed=1    skipped=2    rescued=0    ignored=0
```

- It runs everything correctly until the part in maria db servers and the reason why is because I assign the server 2 which is ubuntu in the db servers and the tasks in the db servers is designed for running and installation of centOS in which we can see that the playbook assigned a condition which is when ansible_distribution is equal to centOS. That is why when the task is restarted, it throws an error since theres no centOS Operating system vm assigned in the db_server and there's no mariadb installed on the server assigned in the db servers.

5. Go to the remote server (Ubuntu) terminal that belongs to the db_servers group and check the status for mariadb installation using the command: systemctl status mariadb. Do this on the CentOS server also. Describe the output.

```
charles-vence-espinas@server2:~$ systemctl status mariadb
Unit mariadb.service could not be found.
```

```
[charles-vence-espinas@centOS ~]$ systemctl status mariadb
Unit mariadb.service could not be found.
```

- The reason why the output is like this because the mariadb is not installed on theserver2 since the play is designed for the installation of centos, not the ubuntu os of server 2. That is why it did not installed.

6. Edit the site.yml again. This time we will append the code to configure installation on the file_servers group. We can add the following on our file.

```
GNU nano 8.4                                site.yml
---

- hosts: all
  become: true
  pre_tasks:

    - name: Install Updates (CentOS)
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: Install updates (Ubuntu)
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  tasks:

    - name: Install apache and php for ubuntu servers
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"
```

```
  - name: install apache and php for CentOS servers
    dnf:
      name:
        - httpd
        - php
      state: latest
    when: ansible_distribution == "CentOS"

- hosts: db_servers
  become: true
  tasks:

  - name: Install mariadb package (CentOS)
    yum:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true

  - name: install mariadb package (Ubuntu)
    apt:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "Ubuntu"
```

```
- hosts: file_servers
  become: true
  tasks:

  - name: install samba package
    package:
      name: samba
      state: latest
```

Make sure to save the file and exit.
Run the site.yml file and describe the result.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml site.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server2]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for
more information.
ok: [centOS]

TASK [Install Updates (CentOS)] ************************************************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] ************************************************************************************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [web_servers] *************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************
ok: [server1]

TASK [Install apache and php for ubuntu servers] *******************************************************************************************************************
ok: [server1]

TASK [install apache and php for CentOS servers] *******************************************************************************************************************
skipping: [server1]

PLAY [db_servers] **************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************
ok: [server2]

TASK [Install mariadb package (CentOS)] ****************************************************************************************************************************
skipping: [server2]

TASK [Mariadb- Restarting/Enabling] ********************************************************************************************************************************
[ERROR]: Task failed: Module failed: Could not find the requested service mariadb: host
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/site.yml:50:7

48       when: ansible_distribution == "CentOS"
49
50     - name: "Mariadb- Restarting/Enabling"
         ^ column 7

fatal: [server2]: FAILED! => {"changed": false, "msg": "Could not find the requested service mariadb: host"}

PLAY RECAP *********************************************************************************************************************************************************
centOS                     : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server2                    : ok=3    changed=0    unreachable=0    failed=1    skipped=2    rescued=0    ignored=0
```

- It did not continue until the end since it throws an error in the maria db and if there are error happened, the remaining play will not occur unless there are multiple hosts in one play, it will still continue the remaining play but only on the active hosts.
  The testing of the file_servers is beyond the scope of this activity, and as well as our topics and objectives. However, in this activity we were able to show that we can target hosts or servers using grouping in ansible playbooks.

**Task 2: Using Tags in running playbooks**

In this task, our goal is to add metadata to our plays so that we can only run the plays that we want to run, and not all the plays in our playbook.

1. Edit the site.yml file. Add tags to the playbook. After the name, we can place the tags: name_of_tag. This is an arbitrary command, which means you can use any name for a tag.

```
---

- hosts: all
  become: true
  pre_tasks:

    - name: Install Updates (CentOS)
      tags: always
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: Install updates (Ubuntu)
      tags: always
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

- hosts: web_servers
  become: true
  tasks:

    - name: Install apache and php for ubuntu servers
      tags: apache, apache2, ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
      when: ansible_distribution == "Ubuntu"
```

```
    - name: install apache and php for CentOS servers
      tags: apache, centos, httpd
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"

- hosts: db_servers
  become: true
  tasks:

    - name: Install mariadb package (CentOS)
      tags: centos, db, mariadb
      yum:
        name: mariadb-server
        state: latest
      when: ansible_distribution == "CentOS"

    - name: "Mariadb- Restarting/Enabling"
      service:
        name: mariadb
        state: restarted
        enabled: true

    - name: install mariadb package (Ubuntu)
      tags: db, mariadb, ubuntu
      apt:
        name: mariadb-server
        state: latest
      when: ansible_distribution == "Ubuntu"
```

```
- hosts: file_servers
  become: true
  tasks:

  - name: install samba package
    tags: samba
    package:
      name: samba
      state: latest
```

Make sure to save the file and exit.
Run the site.yml file and describe the result.

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml site.yml --ask-become-pass
BECOME password:

PLAY [all] ********************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server2]
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server1]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for
more information.
ok: [centOS]

TASK [Install Updates (CentOS)] ***********************************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] ***********************************************************************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [web_servers] ************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************
ok: [server1]
```

```
TASK [Install apache and php for ubuntu servers] ******************************************************************************************************
ok: [server1]

TASK [install apache and php for CentOS servers] ******************************************************************************************************
skipping: [server1]

PLAY [db_servers] *************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************
ok: [server2]

TASK [Install mariadb package (CentOS)] ***************************************************************************************************************
skipping: [server2]

TASK [Mariadb- Restarting/Enabling] *******************************************************************************************************************
[ERROR]: Task failed: Module failed: Could not find the requested service mariadb: host
Origin: /home/charles-vence-espinas/CPE212_CharlesVenceEspinas/ansible-lab/site.yml:55:7

53       when: ansible_distribution == "CentOS"
54
55     - name: "Mariadb- Restarting/Enabling"
         ^ column 7

fatal: [server2]: FAILED! => {"changed": false, "msg": "Could not find the requested service mariadb: host"}

PLAY RECAP ********************************************************************************************************************************************
centOS                     : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server2                    : ok=3    changed=0    unreachable=0    failed=1    skipped=2    rescued=0    ignored=0
```

It also throws an error since same as the earlier, the db server do not have a centos server and that is why

when the playbook tasked to restart the service, it did not run since theres no centos there and theres no mariadb that installed in the host/server.

2. On the local machine, try to issue the following commands and describe each result:

2.1 ansible-playbook --list-tags site.yml

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml --list-tags site.yml

playbook: site.yml

  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [apache, apache2, centos, httpd, ubuntu]

  play #3 (db_servers): db_servers      TAGS: []
      TASK TAGS: [centos, db, mariadb, ubuntu]

  play #4 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
```

2.2 ansible-playbook --tags centos --ask-become-pass site.yml

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml --tags centos --ask-become-pass site.yml
BECOME password:

PLAY [all] ********************************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]

TASK [Install Updates (CentOS)] ***********************************************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] ***********************************************************************************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [web_servers] ************************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************************
ok: [server1]

TASK [install apache and php for CentOS servers] ******************************************************************************************************************
skipping: [server1]
```

```
PLAY [db_servers] *************************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************************
ok: [server2]

TASK [Install mariadb package (CentOS)] ***************************************************************************************************************************
skipping: [server2]

PLAY [file_servers] ***********************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************************
ok: [centOS]

PLAY RECAP ********************************************************************************************************************************************************
centOS                     : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server1                    : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server2                    : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

2.3 ansible-playbook --tags db --ask-become-pass site.yml

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml --tags db --ask-become-pass site.yml
BECOME password:

PLAY [all] ********************************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server2]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for
more information.
ok: [centOS]

TASK [Install Updates (CentOS)] ***********************************************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] ***********************************************************************************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [web_servers] ************************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************************
ok: [server1]

PLAY [db_servers] *************************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************************
ok: [server2]

TASK [Install mariadb package (CentOS)] ***************************************************************************************************************************
skipping: [server2]

TASK [install mariadb package (Ubuntu)] ***************************************************************************************************************************
changed: [server2]

PLAY [file_servers] ***********************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************************
ok: [centOS]

PLAY RECAP ********************************************************************************************************************************************************
centOS                     : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server1                    : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server2                    : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

2.4 ansible-playbook --tags apache --ask-become-pass site.yml

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml --tags apache --ask-become-pass site.yml
BECOME password:

PLAY [all] *********************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]

TASK [Install Updates (CentOS)] ************************************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] ************************************************************************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [web_servers] *************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [server1]

TASK [Install apache and php for ubuntu servers] *******************************************************************************************************
ok: [server1]

TASK [install apache and php for CentOS servers] *******************************************************************************************************
skipping: [server1]

PLAY [db_servers] **************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [server2]

PLAY [file_servers] ************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [centOS]

PLAY RECAP *********************************************************************************************************************************************
centOS                     : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server2                    : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

2.5 ansible-playbook --tags “apache,db” --ask-become-pass site.yml

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml --tags "apache,db" --ask-become-pass si
te.yml
BECOME password:

PLAY [all] ********************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server2]
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server1]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for
more information.
ok: [centOS]

TASK [Install Updates (CentOS)] ***********************************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] ***********************************************************************************************************************
skipping: [centOS]
ok: [server1]
ok: [server2]

PLAY [web_servers] ************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************
ok: [server1]

TASK [Install apache and php for ubuntu servers] ******************************************************************************************************
ok: [server1]
```

```
TASK [install apache and php for CentOS servers] ******************************************************************************************************
skipping: [server1]

PLAY [db_servers] *************************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************
ok: [server2]

TASK [Install mariadb package (CentOS)] ***************************************************************************************************************
skipping: [server2]

TASK [install mariadb package (Ubuntu)] ***************************************************************************************************************
ok: [server2]

PLAY [file_servers] ***********************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************************
ok: [centOS]

PLAY RECAP ********************************************************************************************************************************************
centOS                     : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server2                    : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

## Task 3: Managing Services

1. Edit the file site.yml and add a play that will automatically start the httpd on CentOS server.

```
- name: install apache and php for CentOS servers
  tags: apache, centos, httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

- name: start httpd (CentOS)
  tags: apache, centos, httpd
  service:
    name: httpd
    state: started
  when: ansible_distribution == "CentOS"
```

Figure 3.1.1

Make sure to save the file and exit.

You would also notice from our previous activity that we already created a module that runs a service.

```
- hosts: db_servers
  become: true
  tasks:

    - name: Install mariadb package (CentOS)
      tags: centos, db, mariadb
      yum:
        name: mariadb-server
        state: latest
      when: ansible_distribution == "CentOS"

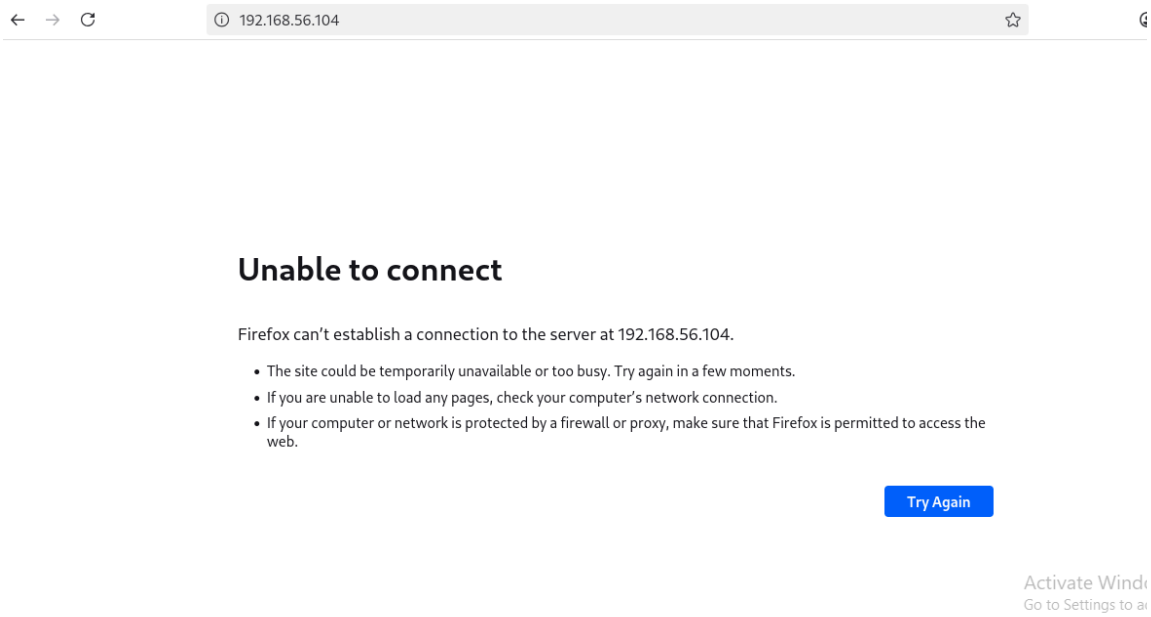
    - name: "Mariadb- Restarting/Enabling"
      service:
        name: mariadb
        state: restarted
        enabled: true
```

Figure 3.1.2

This is because in CentOS, installed packages' services are not run automatically. Thus, we need to create the module to run it automatically.

2. To test it, before you run the saved playbook, go to the CentOS server and stop the currently running httpd using the command sudo systemctl stop httpd. When prompted, enter the sudo password. After that, open the browser and enter the CentOS server's IP address. You should not be getting a display because we stopped the httpd service already.

```
[charles-vence-espinas@centOS ~]$ sudo systemctl stop httpd
[charles-vence-espinas@centOS ~]$
```




3. Go to the local machine and this time, run the site.yml file. Then after running the file, go again to the CentOS server and enter its IP address on the browser.
Describe the result.

```
 GNU nano 8.4                                    inventory.yaml
all:
  vars:
    ansible_user: charles-vence-espinas
  children:

     web_servers:
       hosts:
         server1:
           ansible_host: 192.168.56.102
           os_type: Ubuntu
         centOS:
         ansible_host: 192.168.56.104
         os_type: CentOS

     db_servers:
       hosts:
         server2:
           ansible_host: 192.168.56.103
           os_type: Ubuntu

     file_servers:
       hosts:
         centOS:
           ansible_host: 192.168.56.104
           os_type: CentOS
```

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml site.yml --ask-become-pass
BECOME password:

PLAY [all] *********************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server1]
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server2]
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for
more information.
ok: [centOS]

TASK [Install Updates (CentOS)] ************************************************************************************************************************
skipping: [server1]
skipping: [server2]
ok: [centOS]

TASK [Install updates (Ubuntu)] ************************************************************************************************************************
skipping: [centOS]
ok: [server2]
ok: [server1]

PLAY [web_servers] *************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [server1]
ok: [centOS]

TASK [Install apache and php for ubuntu servers] *******************************************************************************************************
skipping: [centOS]
ok: [server1]

TASK [install apache and php for CentOS servers] *******************************************************************************************************
skipping: [server1]
changed: [centOS]

TASK [start httpd (CentOS)] ****************************************************************************************************************************
skipping: [server1]
changed: [centOS]

PLAY [db_servers] **************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [server2]

TASK [Install mariadb package (CentOS)] ****************************************************************************************************************
skipping: [server2]

TASK [Mariadb- Restarting/Enabling] ********************************************************************************************************************
changed: [server2]

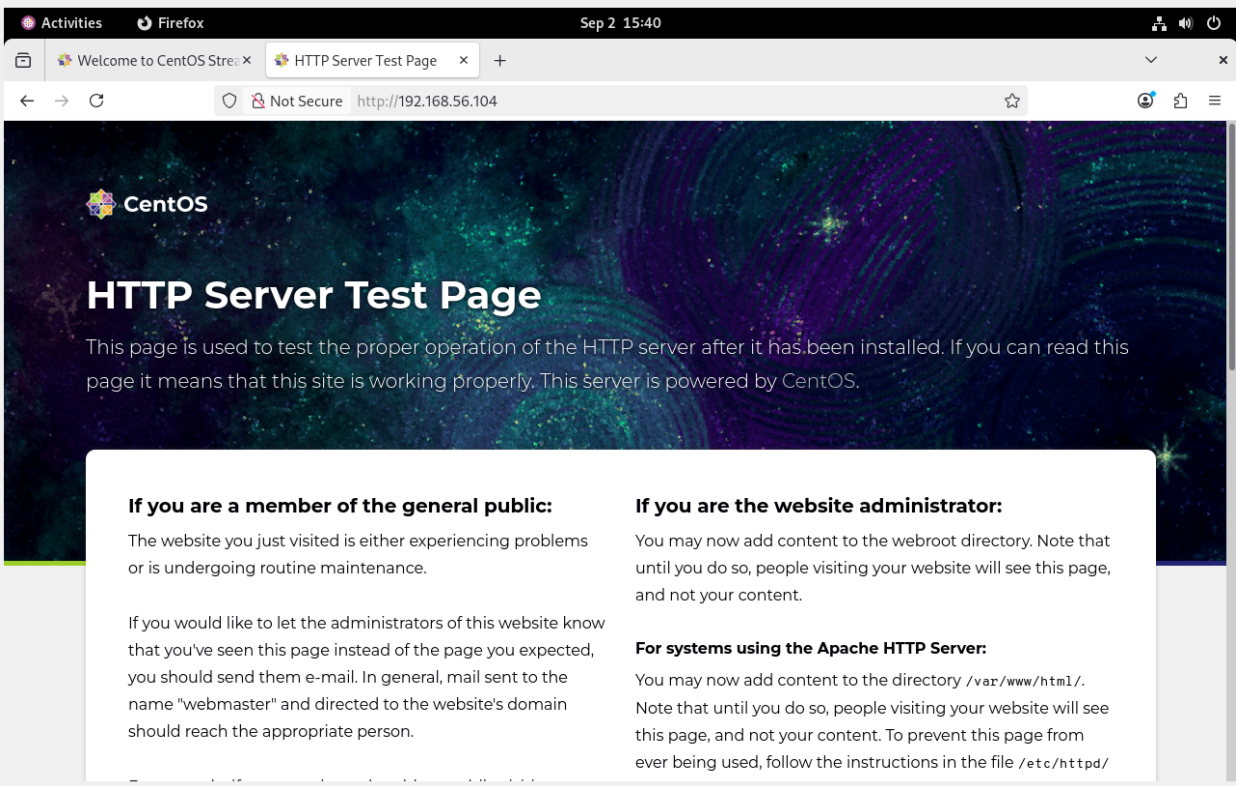
TASK [install mariadb package (Ubuntu)] ****************************************************************************************************************
ok: [server2]

PLAY [file_servers] ************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
ok: [centOS]

TASK [install samba package] ***************************************************************************************************************************
ok: [centOS]

PLAY RECAP *********************************************************************************************************************************************
centOS                     : ok=7    changed=2    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
server1                    : ok=4    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
server2                    : ok=5    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```




- I modify my inventory since I assigned the centos in the files server group which is not a scope of our study so I assign the centOS also to the webservers group so accessing the apache server will be successful. So the output is successful accessing the apache server in centos.

To automatically enable the service every time we run the playbook, use the command enabled: true similar to Figure 7.1.2 and save the playbook.

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

**Supplementary Activity:**

1. Creating a Role for Web Servers
2. Creating a Role for File Servers
3. Creating a Role for Database Servers
4. Combining Multiple Roles in a Playbook

To orchestrate multiple roles in a single playbook for different server groups.
Add tags to each role to allow selective execution, e.g., tags: web for webserver, tags: db for database.

## Web Servers Role

**main.yml in tasks**

```
 GNU nano 8.4                                                  main.yml
---
- name: Install Apache and PHP
  package:
    name:
      - "{{ 'httpd' if ansible_distribution == 'CentOS' else 'apache2' }}"
      - php
    state: present

- name: Start Apache service
  service:
    name: "{{ 'httpd' if ansible_distribution == 'CentOS' else 'apache2' }}"
    state: started
    enabled: true
```

## File Servers Role

**main.yml in tasks**

```
 GNU nano 8.4                                                  main.yml
---
- name: Install Samba
  package:
    name: samba
    state: present

- name: Start Samba Service
  service:
    name: smb
    state: started
    enabled: true
```

## Database Servers Role

**main.yml in tasks**

```
 GNU nano 8.4                                                  main.yml
---
- name: Install MariaDB
  package:
    name: mariadb-server
    state: present

- name: Start MariaDB Service
  service:
    name: mariadb
    state: started
    enabled: true
```

## Inventory

```
 GNU nano 8.4                                                inventory.yaml
all:
  vars:
    ansible_user: charles-vence-espinas
  children:

      web_servers:
        hosts:
          server1:
            ansible_host: 192.168.56.102
            os_type: Ubuntu

      db_servers:
        hosts:
          server2:
            ansible_host: 192.168.56.103
            os_type: Ubuntu

      file_servers:
        hosts:
          centOS:
            ansible_host: 192.168.56.104
            os_type: CentOS
```

## Playbook

```
 GNU nano 8.4                                          supplementary6.yml
---
- name: Configure Web Servers
  hosts: web_servers
  become: true

  roles:
    - role: web_servers
      tags:
        - web

- name: Configure File Servers
  hosts: file_servers
  become: true

  roles:
    - role: file_servers
      tags:
        - file

- name: Configure Database Servers
  hosts: db_servers
  become: true

  roles:
    - role: db_servers
      tags:
        - db
```

## Run all

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml supplementary6.yml --ask-become-pass
BECOME password:

PLAY [Configure Web Servers] ***************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server1]

TASK [web_servers : Install Apache and PHP] ************************************************************************************************************
changed: [server1]

TASK [web_servers : Start Apache service] **************************************************************************************************************
ok: [server1]

PLAY [Configure File Servers] **************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for
more information.
ok: [centOS]

TASK [file_servers : Install Samba] ********************************************************************************************************************
ok: [centOS]

TASK [file_servers : Start Samba Service] **************************************************************************************************************
changed: [centOS]
```

```
PLAY [Configure Database Servers] **********************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server2]

TASK [db_servers : Install MariaDB] ********************************************************************************************************************
ok: [server2]

TASK [db_servers : Start MariaDB Service] **************************************************************************************************************
ok: [server2]

PLAY RECAP *********************************************************************************************************************************************
centOS                     : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

## Run only web servers

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml supplementary6.yml --tags web --ask-bec
ome-pass
BECOME password:

PLAY [Configure Web Servers] ***************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server1]

TASK [web_servers : Install Apache and PHP] ************************************************************************************************************
ok: [server1]

TASK [web_servers : Start Apache service] **************************************************************************************************************
ok: [server1]

PLAY [Configure File Servers] **************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for
more information.
ok: [centOS]

PLAY [Configure Database Servers] **********************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server2]

PLAY RECAP *********************************************************************************************************************************************
centOS                     : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

## Run only file servers

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml supplementary6.yml --tags file --ask-be
come-pass
BECOME password:

PLAY [Configure Web Servers] ***************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server1]

PLAY [Configure File Servers] **************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter cou
ld cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for
more information.
ok: [centOS]

TASK [file_servers : Install Samba] ********************************************************************************************************************
ok: [centOS]

TASK [file_servers : Start Samba Service] **************************************************************************************************************
ok: [centOS]

PLAY [Configure Database Servers] **********************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter c
ould cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html fo
r more information.
ok: [server2]

PLAY RECAP *********************************************************************************************************************************************
centOS                     : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

**Run only database servers**

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab$ ansible-playbook -i inventory.yaml supplementary6.yml --tags db --ask-become-pass
BECOME password:

PLAY [Configure Web Servers] ***************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'server1' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

PLAY [Configure File Servers] **************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'centOS' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [centOS]

PLAY [Configure Database Servers] **********************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************
[WARNING]: Host 'server2' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]

TASK [db_servers : Install MariaDB] ********************************************************************************************************************
ok: [server2]

TASK [db_servers : Start MariaDB Service] **************************************************************************************************************
ok: [server2]

PLAY RECAP *********************************************************************************************************************************************
centOS                     : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server1                    : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

**Reflections:**

Answer the following:

1. What is the importance of putting our remote servers into groups?
   - The importance of putting the remote servers into groups is that it makes the server easier to manage since we can group them based on their purpose like the web servers, file servers and database servers and run specific tasks only on the servers that need them.
2. What is the importance of tags in playbooks?
   - The importance of tags in the playbooks are that they allow us to choose which tasks or roles we want to run. Instead of running the whole playbook, we can use a specific tag to run only the needed tasks.
3. Why do think some services need to be managed automatically in playbooks?
   - Some services need to managed automatically in the playbooks because it saves time and make sure that the services are running properly. It is easier than manually starting or configuring the services on every remote servers.

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **Gemini**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **Explanation of concepts and how the code works provided by the procedure**

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*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **I use this tool to help me understand deeply the concept of roles and tags and the code provided in the procedure**

*Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.*

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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